

Learning Guide Unit 5

Overview

UNIT 5: Advanced Tree Structures and Recursion Techniques

Topics

- Balanced Binary Search Trees (AVL Trees, Red-Black Trees).
 - Binary Heap Operations
 - Recursive Tree Algorithms
 - Dynamic Programming and Recursion
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Learning Objectives

By the end of this unit, you will be able to:

1. Describe the operations and properties of AVL Trees and Red-Black Trees.
2. Implement binary heap operations to solve real-world problems.
3. Apply recursive algorithms to perform tree operations.
4. Solve problems efficiently using dynamic programming.